SECTIONS OF A CONSTANT TENSOR PRODUCT: \UNDERLINE{M \OTIMES_\LAMBDA Q}(U) = \UNDERLINE{M}(U) \OTIMES_\LAMBDA Q FOR FINITELY PRESENTED Q

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one same-chapter lemma (lemma-constant-exact, already carded in this lane), recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Exactness of the constant sheaf functor on exact sequences. Stacks Project, Modules on Sites, Lemma lemma-constant-exact (tag 01AC-series; carded locally as S11-sites-mod-constant-exact). Supplies that applying \underline{\ } to the tensored exact sequence yields an exact sequence of section groups.
- Right exactness of tensor product. Stacks Project, Algebra, section on tensor products. Underlies the step producing M^{\oplus m} \to M^{\oplus n} \to M \otimes_\Lambda Q \to 0 from the presentation of Q.

Source reference. Original lemma 093J: sources/stacks-chow/tags.txt, source lines 11517–11517. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let Λ be a ring and let M and Q be Λ-modules. If Q is a finitely presented Λ-module, then we have $\underline{M \otimes_\Lambda Q}(U) = \underline{M}(U) \otimes_\Lambda Q$ for all $U \in \mathrm{Ob}(\mathcal{C})$.*

Proof. Choose a presentation $\Lambda^{\oplus m} \to \Lambda^{\oplus n} \to Q \to 0$. This gives an exact sequence $M^{\oplus m} \to M^{\oplus n} \to M \otimes Q \to 0$. By Lemma 093J we obtain an exact sequence

$$\underline{M}(U)^{\oplus m} \to \underline{M}(U)^{\oplus n} \to \underline{M \otimes Q}(U) \to 0$$

which proves the lemma. (Note that taking sections over U always commutes with finite direct sums, but not arbitrary direct sums.) □

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